

Lean and LeanAlde

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September 19, 2024

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Some stories

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- In 1937, a German-Jewish mathematician Samuel I. Krieger once claimed to have found a counterexample to Fermat’s last theorem : $x^n + y^n = z^n$, for $x = 1324$, $y = 731$, and $z = 1961$. Note that the LHS can only end in 5 or 7, while the RHS can end in 1.

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- Chapter 3, What is Mathematics, Really? : “Newton found the speed of a falling stone by dividing $0/0$. Berkeley called him to account for bad algebra, but admitted Newton had the right answer... Mistake or not?”

Excerpt from What is Mathematics, Really?

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Henri Poincaré said it was strange that mistakes happen in mathematics, since mathematics is just sound reasoning, such as anyone in his right mind follows. His explanation was memory lapse there are only so many things we can keep in mind at once. .. Some mistakes come from keeping old assumptions in a new context. .. We mathematicians make mistakes, even important ones, even in famous papers that have been around for years.

Excerpt from What is Mathematics, Really?

Can we be sure that we, unlike our predecessors, are not overlooking big gaps?

The question

Can we find a reliable and uniform system to validate the precision of the mathematics we create?

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- The guiding spirit of `mathlib` is to prove theorems in the highest possible generality.
- `mathlib` uses type theory as a building block, instead of set theory, which leads to subtle (but consequential) differences in proofs on paper and in Lean.

A little bit about type theory

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- The type of propositions, Prop , is special : a term of type a proposition p is a proof of it.
- Lean does not remember the proof, just its existence.
- Let us take a look at an example.

Some theories that have been formalized in Lean

- Broadly all undergraduate math curricula
- Perfectoid spaces (Kevin Buzzard, Johan Commelin, Patrick Massot, 2020)
- Gardam's disproof of the unit conjecture (Siddhartha Gadgil, Anand Rao Tadipatri, 2022)
- Condensed mathematics (Johan Commelin et al, 2022)
- Independence of the continuum hypothesis (Jesse Han, Floris van Doorn)
- Existence of sphere eversions (Patrick Massot, Floris van Doorn, Oliver Nash, 2021)

Some number theory results that have been formalized in Lean

- Quadratic reciprocity (Michael Stoll)
- Pell's equation and Matiyasevic's theorem (Mario Carneiro, Michael Stoll)
- Hensel's lemma on $\mathbb{Z}_p$ (Rob Lewis)
- Finiteness of the class number (Anne Baanen, Sander Dahmen, Filippo Nuccio, A.N.)
- Dirichlet's Unit Theorem (Xavier Roblot)
- Adèles and idéles (María Inés de Frutos-Fernández, 2022)
- Local fields and class field theory (María Inés de Frutos-Fernández, Filippo A.E. Nuccio et al)
- p -adic L -functions (A.N.)
- Fermat's last theorem? (Kevin Buzzard et al.)

What are p -adic L -functions?

Kummer's congruence states that, when h and k are positive even integers not divisible by $p - 1$:

$$\frac{B_h}{h} \equiv \frac{B_k}{k} \pmod{p}$$

for the Bernoulli numbers B_h and B_k . Kubota and Leopoldt's search for a meromorphic function that helps study this congruence led them to the p -adic L -function. They are characterized by their values at negative integers,

$$L_p(1 - n, \chi) = -(1 - \chi\omega^{-n}(p)p^{n-1}) \frac{B_{n, \chi\omega^{-n}}}{n}$$

for $n > 1$.

We are going to define p -adic L -functions in terms of a p -adic integral.